

This query retrieves all rows in the EMPLOYEES table, even if there is no match in the DEPARTMENTS table. It also retrieves all rows in the DEPARTMENTS table, even if there is no match in the EMPLOYEES table.

Find the Solution for the following:

1. Write a query to display the last name, department number, and department name for all employees.

```
SELECT e.last_name, e.department_id, d.department_name
FROM employees e
JOIN departments d ON e.department_id = d.department_id;
```

2. Create a unique listing of all jobs that are in department 80. Include the location of the department in the output.

```
SELECT DISTINCT e.job_id, d.location_id
FROM employees e
JOIN departments d ON e.department_id = d.department_id
WHERE e.department_id = 80;
```

3. Write a query to display the employee last name, department name, location ID, and city of all employees who earn a commission

```
SELECT e.last_name, d.department_name, d.location_id, l.city
FROM employees e JOIN departments d ON e.department_id =
d.department_id JOIN locations l ON d.location_id =
l.location_id WHERE e.commission_pct IS NOT NULL;
```

8. Display the employee last name and department name for all employees who have an (lowercase) 'a' in their last names.

```
SELECT e.last_name, d.department_name
FROM employees e JOIN departments d ON e.department_id =
d.department_id WHERE e.last_name LIKE '%a%';
```

5. Write a query to display the last name, job, department number, and department name for all employees who work in Toronto.

```
SELECT e.last_name, e.job_id,
e.department_id, d.department_name
FROM employees e JOIN departments d ON e.department_id = d.department_id
JOIN locations l ON d.location_id = l.location_id
WHERE l.city = 'Toronto';
```

6. Display the employee last name and employee number along with their manager's last name and manager number. Label the columns Employee, Emp#, Manager, and Mgr#, respectively

```
SELECT w.last_name AS "Employee", w.employee_id AS
"Emp #", m.last_name AS "Manager", m.employee_id AS
"Mgr #" FROM employees w JOIN employees m
ON w.manager_id = m.employee_id;
```

7. Modify lab4_6.sql to display all employees including King, who has no manager. Order the results by the employee number.

```
SELECT w.last_name AS "Employee", w.employee_id AS  
"Emp #", m.last_name AS "Manager", m.employee_id AS  
"Mgr #", FROM employees w LEFT OUTER JOIN employees m  
ON w.manager_id = m.employee_id ORDER BY w.employee_id
```

8. Create a query that displays employee last names, department numbers, and all the employees who work in the same department as a given employee. Give each column an appropriate label

```
SELECT e1.department_id, e1.last_name AS "Employee", e2.last_name AS  
"Colleague" FROM employees e1 JOIN employees e2 ON e1.  
JOIN department_id = e2.department_id WHERE  
e1.last_name = 'a employee_name' AND e1.employee_id  
< e2.employee_id
```

9. Show the structure of the JOB_GRADES table. Create a query that displays the name, job, department name, salary, and grade for all employees

```
SELECT e.last_name, j.job_id, d.department_name,  
e.salary, j.grade_level FROM employees e JOIN departments  
ON e.department_id = d.department_id JOIN job_grades g  
ON e.job_id = g.job_id AND  
e.salary BETWEEN g.lowest_sal AND  
g.highest_sal
```

10. Create a query to display the name and hire date of any employee hired after employee Davies.

```
SELECT last_name, hire_date FROM employees  
WHERE hire_date > (SELECT hire_date FROM  
employees WHERE last_name = 'Davies');
```

11. Display the names and hire dates for all employees who were hired before their managers, along with their manager's names and hire dates. Label the columns Employee, Emp Hired, Manager, and Mgr Hired, respectively.

```
SELECT w.last_name AS "Employee", w.hire_date  
AS "Emp Hired", m.last_name AS "Manager", m.hire_date  
AS "Mgr Hired" FROM employees w JOIN  
employees m ON w.manager_id = m.employee_id  
WHERE w.hire_date < m.hire_date;
```

Evaluation Procedure	Marks awarded
Query(5)	5
Execution (5)	5
Viva(5)	5
Total (15)	15
Faculty Signature	